QUESTION 7: HDFS File Management – Deleting Files from HDFS

Aim

To perform file management operations in Hadoop Distributed File System (HDFS) by deleting files and directories using Hadoop commands and verifying the deletion process.

Procedure

1. First, ensure that Hadoop is properly installed and configured in the system.
2. Open the terminal and start the Hadoop Distributed File System using:

start-dfs.sh

3. Verify whether the Hadoop services are running using the `jps` command.

jps

4. Create a directory in HDFS to store sample files.

```
hdfs dfs -mkdir -p /user/student/input
```

5. Create a sample file in the local file system.

echo "This is a sample file for HDFS deletion" > data.txt

6. Upload the file into the HDFS directory using the `-put` command.

```
hdfs dfs -put data.txt /user/student/input/
```

7. Verify whether the file is available in HDFS by listing the directory contents.

```
hdfs dfs -ls /user/student/input
```

8. To delete a specific file from HDFS, use the `-rm` command followed by the complete file path.

```
hdfs dfs -rm /user/student/input/data.txt
```

9. The command removes the specified file from the HDFS namespace. Verify the deletion by listing the directory again.

```
hdfs dfs -ls /user/student/input
```

10. To delete an entire directory along with all its contents, use the `-rm -r` command.

```
hdfs dfs -rm -r /user/student/old_directory
```

11. To delete a directory only if it is empty, use the `-rmdir` command.

```
hdfs dfs -rmdir /user/student/empty_directory
```

12. To check whether the file or directory still exists in HDFS, use the `-test` command.

```
hdfs dfs -test -e /user/student/input/data.txt
```

echo $?

If the file does not exist, the test command returns a non-zero exit status.

13. Thus, the file and directory deletion operations are successfully performed and verified in HDFS.

Program

HDFS file deletion commands:

Start Hadoop services

start-dfs.sh

Verify Hadoop services

jps

Create a directory in HDFS

```
hdfs dfs -mkdir -p /user/student/input
```

Create a sample local file

echo "This is a sample file for HDFS deletion" > data.txt

Upload the file into HDFS

```
hdfs dfs -put data.txt /user/student/input/
```

Display files before deletion

```
hdfs dfs -ls /user/student/input
```

Delete a specific file from HDFS

```
hdfs dfs -rm /user/student/input/data.txt
```

Verify file deletion

```
hdfs dfs -ls /user/student/input
```

Create a sample directory

```
hdfs dfs -mkdir -p /user/student/old_directory
```

Delete the directory recursively

```
hdfs dfs -rm -r /user/student/old_directory
```

Verify directory deletion

```
hdfs dfs -ls /user/student
```

Output of Program

1. Hadoop service verification

```
$ jps
```

```
NameNode
```

DataNode

```
SecondaryNameNode
```

Jps

2. Displaying files before deletion

```
$ hdfs dfs -ls /user/student/input
```

```
Found 1 items
```

-rw-r--r-- 1 student supergroup 40

/user/student/input/data.txt

3. Deleting a file

```
$ hdfs dfs -rm /user/student/input/data.txt
```

```
Deleted /user/student/input/data.txt
```

4. Verifying file deletion

```
$ hdfs dfs -ls /user/student/input
```

No files found.

5. Deleting a directory

```
$ hdfs dfs -rm -r /user/student/old_directory
```

Deleted /user/student/old_directory

6. Verifying directory deletion

```
$ hdfs dfs -ls /user/student
Found 1 items
```

/user/student/input

Result

Thus, the HDFS file management and deletion operations were successfully performed. A sample file was created and uploaded into HDFS, and the file was deleted using the `-rm` command. An entire directory was also deleted recursively using the `-rm -r` command. The deletion operations were verified successfully using the Hadoop file system listing commands.